Session 3 — Feature branches and Pull Requests

Shared topic (Python FS + Java FS) · **~75 minutes**

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Session Outcome

`main` is the copy you would demo. Login is not finished. You will work on a **separate line**, push it, and ask GitHub to take it into `main` only after a look.

By the end you will be able to:

- Say why unfinished login must not sit on `main`
- Create, switch, and list branches
- Commit on a feature branch and push that branch
- Open a Pull Request, merge it, then update local `main`
- Use a PR even when you are the only developer

This course uses one pattern: **`main` + feature branch + Pull Request**. We will not learn extra branching strategies.

Conflicts stay in Session 4. If Git cannot combine two branches automatically today, stop and wait.

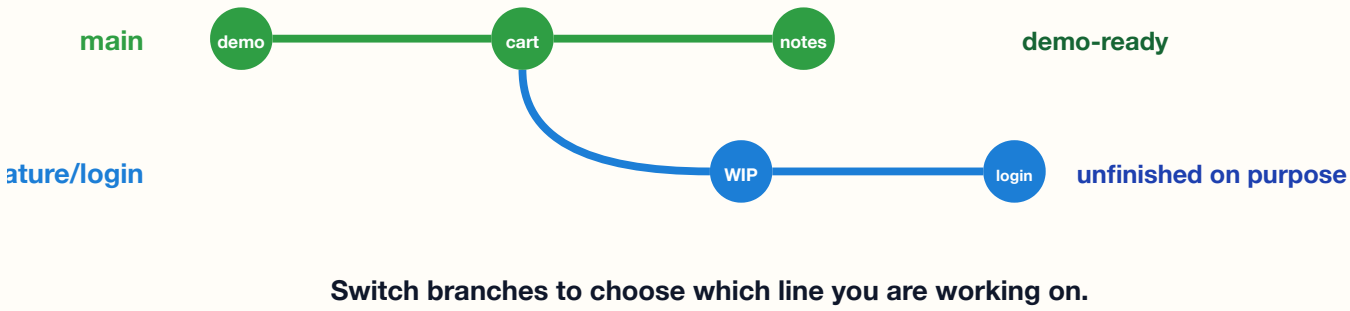
1. The problem

KSR_MART on `main` works. A teammate (or a trainer) can clone it and run the demo.

You need a **login** page. It will take two days. If you commit half-built login on `main`, the demo is broken until you finish.

Need: an isolated line of work. `main` stays demo-ready. Login lives somewhere else until it is ready.

That isolated line is a **branch**.



`main` is just the name of the stable line. A **feature branch** is a line you create for one job — login, payment, a bug fix.

You switch branches to choose which line you are editing. Files on disk follow the branch you are on.

2. Work on a feature branch

Use `~/KSR_MART` (not the clone from Session 2, unless you prefer one folder and stay in it).

First, stand on the latest `main`:

```
cd ~/KSR_MART
git switch main
git pull
```

Create a branch **and** move onto it:

```
git switch -c feature/login
```

`-c` means create. `feature/login` is a name, not a special Git type. Teams often use `feature/...` so the job is obvious.

Check where you are:

```
git branch
```

The star is the current branch. You want `* feature/login`.

Now the weekday loop from Session 2 — it belongs to this branch, not to `main`:

```
echo "# Login" > LOGIN.md
echo "WIP: email and password form." >> LOGIN.md
git add LOGIN.md
git commit -m "Add login notes"
```

Switch back to `main` without merging:

```
git switch main
```

`LOGIN.md` should **disappear** from the folder. It is not gone. It lives on `feature/login`. Switch back:

```
git switch feature/login
```

The file returns. That is the point of a branch.

List branches again any time you feel lost: `git branch`. Older notes use `git checkout`. This course uses `git switch`.

3. Push the branch, not only `main`

GitHub does not see `feature/login` until you push **that** branch.

```
git switch feature/login
git push -u origin feature/login
```

`-u` remembers the link, same idea as Session 1. After that, `git push` on this branch is enough.

On GitHub, open the repo. You should see a banner that `feature/login` had a recent push, and a **Compare & pull request** button.

Do **not** merge on the laptop for this course. The team habit is: push the branch, then open a Pull Request.

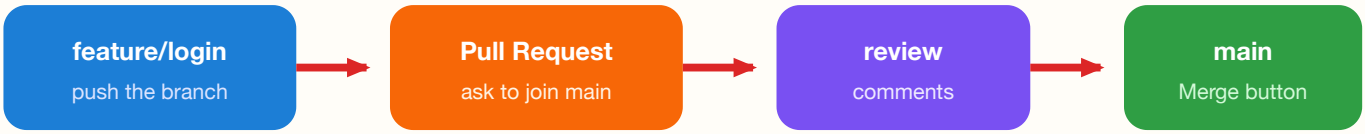
4. What a Pull Request is

A **Pull Request (PR)** is a GitHub page that says:

> Please take the commits on `feature/login` into `main`.

It is not a Git command. It is a request on the website.

Need: someone looks at the change *before* it becomes the demo copy.



On GitHub:

1. **Compare & pull request** (or **Pull requests → New**)
2. Base: `main` ← compare: `feature/login`
3. Title like a commit message: `Add login notes`
4. Short description: what you did, how to check it
5. **Create pull request**

A teammate can comment, ask for a change, or approve.

Then **Merge pull request** → confirm. Tick **Delete branch** if GitHub offers it (that is the GitHub copy of `feature/login`).

`main` on GitHub now has login. Your laptop `main` does not until you pull.

```
git switch main
git pull
```

`LOGIN.md` should appear on `main`. The local feature branch can go:

```
git branch -d feature/login
```

`-d` refuses if the work never made it into `main`. Do not use `-D` unless you mean to throw the branch away.

5. Solo vs team

Team: Aisha opens a PR. Rohan reviews. After approval, someone merges. Unfinished login never lands on `main` by a quiet `git push` to `main`.

Solo: You still open a PR on your own `ksr-mart`. You are the reviewer. You read the diff on GitHub, then merge yourself.

A PR is useful even for one person:

- You see the change in one place before it is "the project"
- A recruiter can see you know the team habit
- You can add a second commit on `feature/login` (`git add`, `commit`, `git push`) and the same PR updates

Do not push straight to `main` on a team repo. On your own portfolio, practise the PR anyway so the buttons are not new in an internship.

6. Try this — one PR for login

```
cd ~/KSR_MART
git switch main
git pull
git switch -c feature/login
echo "# Login" > LOGIN.md
echo "WIP: email and password form." >> LOGIN.md
git add LOGIN.md
git commit -m "Add login notes"
git push -u origin feature/login
```

On GitHub: open the PR → merge → delete the remote branch.

```
git switch main
git pull
git branch -d feature/login
git log --oneline -5
```

You should see `Add login notes` on `main`. `git branch` should no longer list `feature/login`.

Optional second pass: `feature/payment` with one line in `PAYMENT.md`, same PR flow. Aisha login / Rohan payment is the two-laptop story from Session 1, done the professional way.

Common mistakes

Every edit on `main`. One bad save breaks the demo. New work starts with `git switch -c feature/...` from an up-to-date `main`.

Forgot to push the branch. The PR page is empty or GitHub has no `feature/login`. `git push -u origin feature/login`.

PR from `main` into `main`. Nothing useful to review. Compare the **feature** branch into `main`.

Deleted the branch before the PR was merged. The work may still be on GitHub if you pushed. If you never pushed, the commits may exist only on this laptop. Push first. Merge first. Delete last.

Laptop `main` looks old after a GitHub merge. You merged on the website. `git switch main` then `git pull`.

`git switch` fails: uncommitted changes. Finish or stash is Session 4. For today: `status`, then `add` / `commit` on the current branch before you switch — or copy the file aside and ask the trainer.

Key takeaway

```
main (stable)
   ↓
feature/login  →  commit  →  push  →  Pull Request  →  review  →  merge
   ↓
git switch main && git pull
```

A branch is an isolated line. A Pull Request is how that line asks to join `main`.

Session 4: a teammate already pushed `main`, two people edited the same file, or you committed the wrong thing. That sitting is sync, safe undo, and one merge conflict.